

Statistical Data Visualization

Method Dossier | Basic Level

1. Method Description

Good visualization reveals patterns invisible in summary statistics (Anscombe quartet).

Distributions: histogram (Sturges rule), boxplot, violin plot, Q-Q plot.

Relationships: scatterplot + regression line, scatterplot matrix. Handle overplotting with jitter/alpha/2D-histogram.

Group comparisons: raincloud plot (violin+boxplot+stripchart). Bar charts with error bars (less informative).

Time series: line plot with CI band, spaghetti plot, slope plot.

2. Field-Specific Examples

Medicine: forest plot (meta-analysis), Kaplan-Meier curve with risk table, Bland-Altman plot.

Psychology: raincloud plot for group comparison, Q-Q plot, scatterplot with regression.

Economics: correlation heatmap, waterfall chart, time series with moving average.

3. Principles

y-axis at 0 (except log/difference). No 3D (distorts proportions).

Color: colorblind-friendly (ColorBrewer, viridis). Maximize data-ink ratio (Tufte).

4. Code Examples

R:

```
ggplot(df, aes(x=gruppe, y=wert)) + geom_violin() + geom_boxplot(width=0.2)
ggplot(df, aes(x=x, y=y)) + geom_point(alpha=0.5) + geom_smooth(method="lm")
ggsurvplot(fit); corrplot(cor(df))
```

Python:

```
sns.violinplot(data=df, x='gruppe', y='wert')
sns.regplot(data=df, x='x', y='y')
```

5. Exercises

- Two groups (n=50), metric DV. Best plot type? Create in R/Python. What info does bar chart with error bars miss?
- Scatterplot of 500 points has overplotting. Three methods to handle it. Why is 3D bar chart problematic?